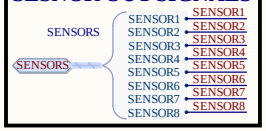
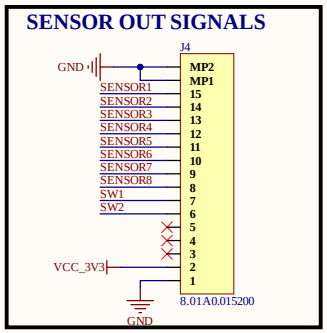
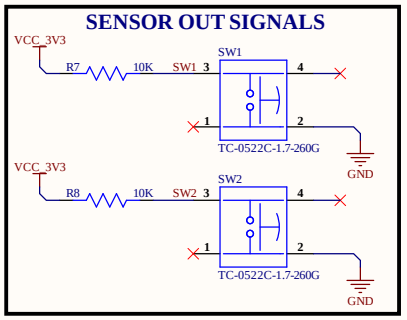
SESNOR OUT SIGNALS



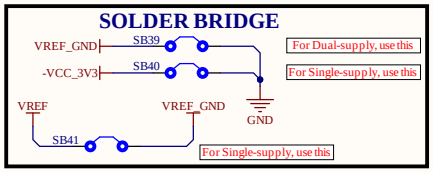
SENSOR OUT SIGNALS



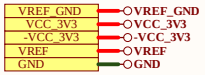
SENSOR OUT SIGNALS



SOLDER BRIDGE



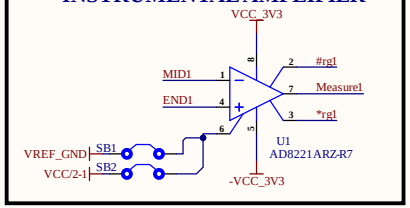
For Single-supply, solder SB3, SB5, SB6
For Double-supply, solder SB1, SB2, SB4



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Title: Connector-Sheet.SchDoc	
File: EMG-Arm-ELEC-V2.0.PrjPcb	
Date: 11/02/2025	Page Num: *
Drawn By: Huzaifa Ali	Page Total: 11

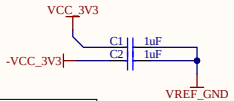
INSTRUMENTAL AMPLIFIER



VREF input requires almost 50uA current. This would make the total current of 8 opamps equal to 400uA.

Supporting Component

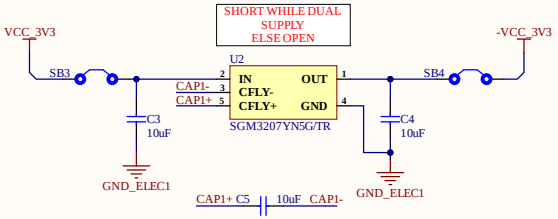
DECOUPLING CAPS



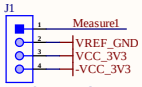
Place these near AD8221ARZ-R7

DUAL SUPPLY IC

SHORT WHILE DUAL SUPPLY ELSE OPEN

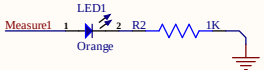


TEST PADS

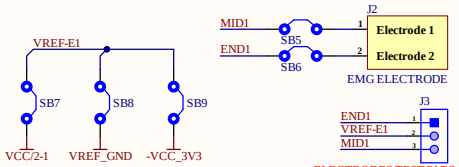


4 Pin SMD Wire Connector

LED

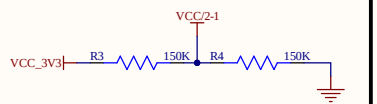


sEMG SENSOR

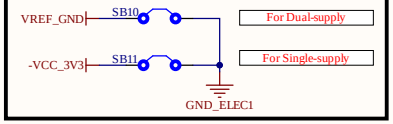


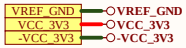
For Dual-supply, short VREF_GND SB

Voltage Divider



Solder Bridges



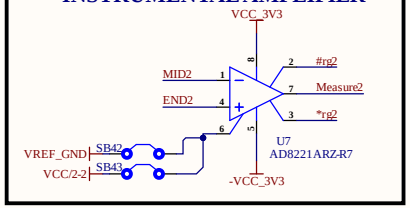




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File: EMG-Arm-ELEC-V2.0.PrjPcb		
Date: 08/02/2025	Page	Num: *
Drawn By: Huzaifa Ali	Page	Total: 11

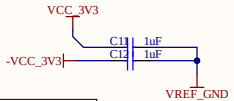
INSTRUMENTAL AMPLIFIER



VREF input requires almost 50uA current. This would make the total current of 8 opamps equal to 400uA.

Supporting Component

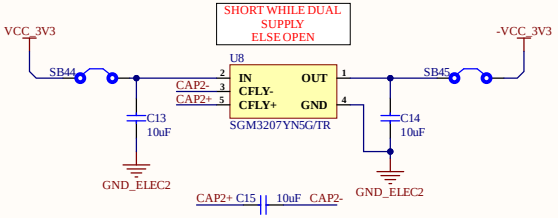
DECOUPLING CAPS



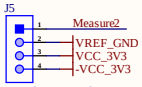
Place these near AD8221ARZ-R7

DUAL SUPPLY IC

SHORT WHILE DUAL SUPPLY ELSE OPEN

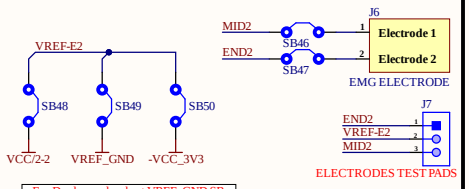


TEST PADS



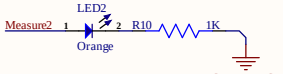
4 Pin SMD Wire Connector

sEMG SENSOR

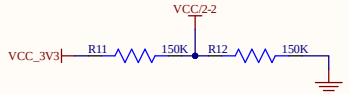


For Dual-supply, short VREF_GND SB

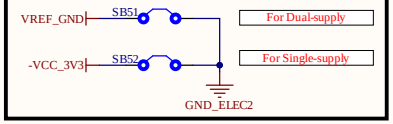
LED

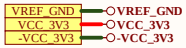


Voltage Divider



Solder Bridges



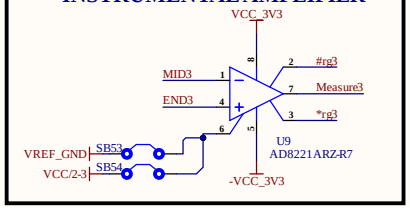




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Title: Electrode_Sheet2.SchDoc		
File: EMG-Arm-ELEC-V2.0.PrjPcb		
Date: 08/02/2025	Page	Num: *
Drawn By: Huzaifa Ali	Page	Total: 11

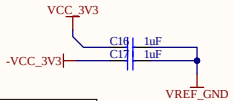
INSTRUMENTAL AMPLIFIER



VREF input requires almost 50uA current. This would make the total current of 8 opamps equal to 400uA.

Supporting Component

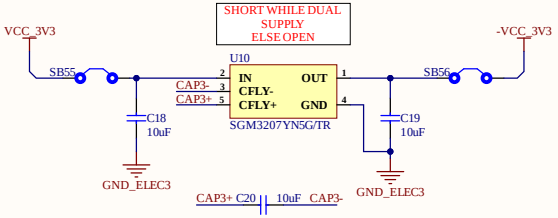
DECOUPLING CAPS



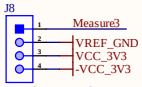
Place these near AD8221ARZ-R7

DUAL SUPPLY IC

SHORT WHILE DUAL SUPPLY ELSE OPEN

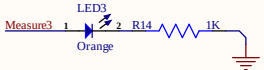


TEST PADS

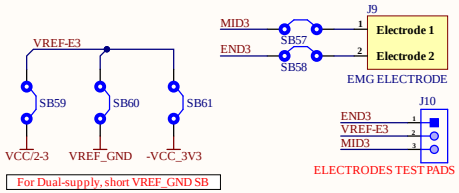


4 Pin SMD Wire Connector

LED

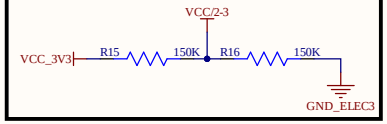


sEMG SENSOR

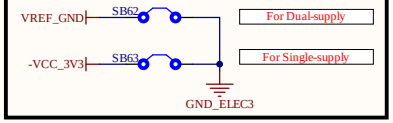


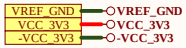
For Dual-supply, short VREF_GND SB

Voltage Divider



Solder Bridges



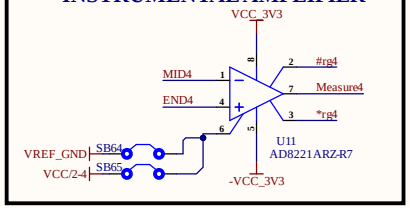




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Date: 08/02/2025	Page	Num: *
Drawn By: Huzaifa Ali	Page	Total: 11

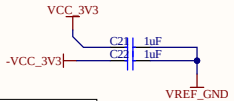
INSTRUMENTAL AMPLIFIER



VREF input requires almost 50uA current. This would make the total current of 8 opamps equal to 400uA.

Supporting Component

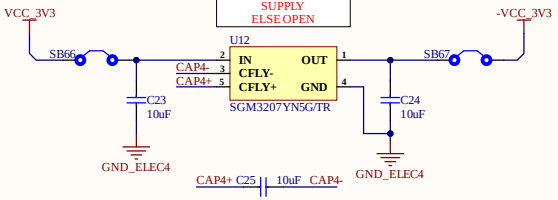
DECOUPLING CAPS



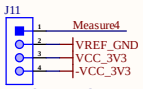
Place these near AD8221ARZ-R7

DUAL SUPPLY IC

SHORT WHILE DUAL SUPPLY ELSE OPEN

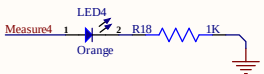


TEST PADS

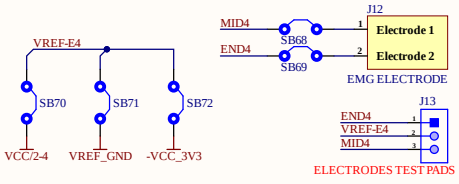


4 Pin SMD Wire Connector

LED

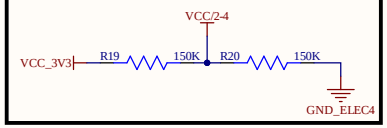


sEMG SENSOR

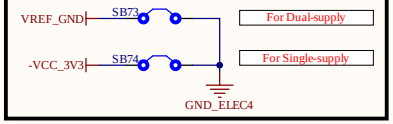


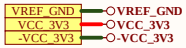
For Dual-supply, short VREF_GND SB

Voltage Divider



Solder Bridges



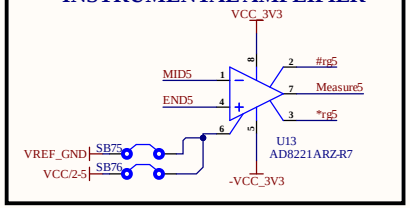




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Title: Electrode_Sheet4.SchDoc		
File: EMG-Arm-ELEC-V2.0.PrjPcb		
Date: 08/02/2025	Page	Num: *
Drawn By: Huzaifa Ali	Page	Total: 11

INSTRUMENTAL AMPLIFIER

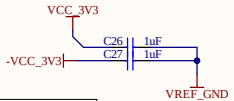


VREF input requires almost 50uA current. This would make the total current of 8 opamps equal to 400uA.

Supporting Component



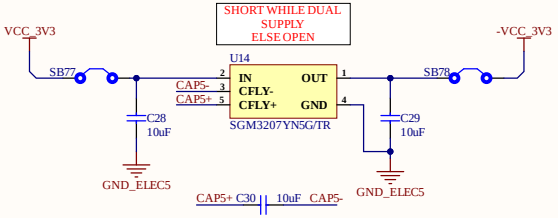
DECOUPLING CAPS



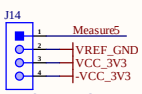
Place these near AD8221ARZ-R7

DUAL SUPPLY IC

SHORT WHILE DUAL SUPPLY ELSE OPEN

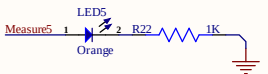


TEST PADS

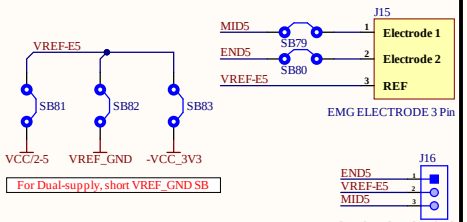


4 Pin SMD Wire Connector

LED

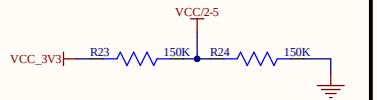


sEMG SENSOR

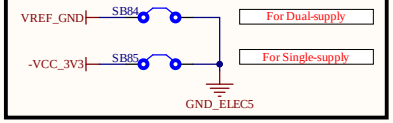


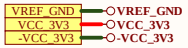
For Dual-supply, short VREF_GND SB

Voltage Divider



Solder Bridges



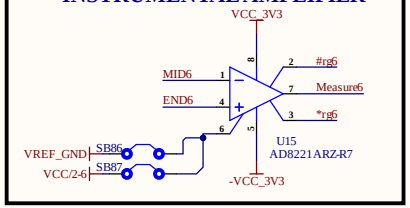




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Title: Electrode_Sheet5.SchDoc		
File: EMG-Arm-ELEC-V2.0.PrjPcb		
Date: 08/02/2025	Page	Num: *
Drawn By: Huzaifa Ali	Page	Total: 11

INSTRUMENTAL AMPLIFIER

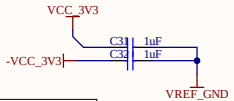


VREF input requires almost 50uA current. This would make the total current of 8 opamps equal to 400uA.

Supporting Component



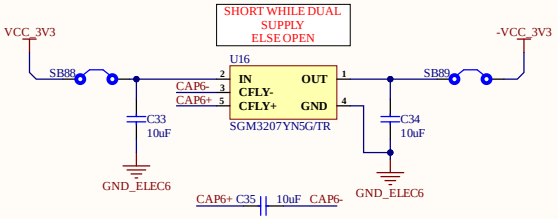
DECOUPLING CAPS



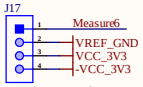
Place these near AD8221ARZ-R7

DUAL SUPPLY IC

SHORT WHILE DUAL SUPPLY ELSE OPEN

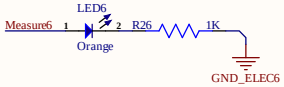


TEST PADS

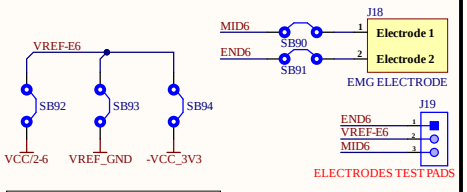


4 Pin SMD Wire Connector

LED

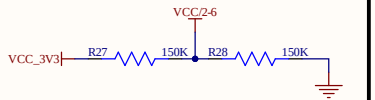


sEMG SENSOR

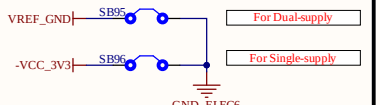


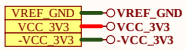
For Dual-supply, short VREF_GND SB

Voltage Divider



Solder Bridges



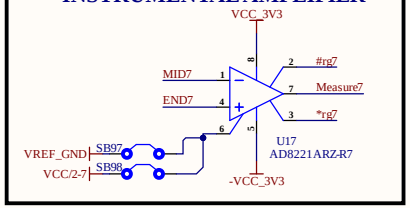




eSpark Consulting Group

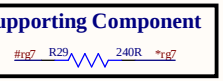
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File: EMG-Arm-ELEC-V2.0.PrjPcb		
Date: 08/02/2025	Page	Num: *
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INSTRUMENTAL AMPLIFIER

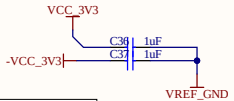


VREF input requires almost 50uA current. This would make the total current of 8 opamps equal to 400uA.

Supporting Component

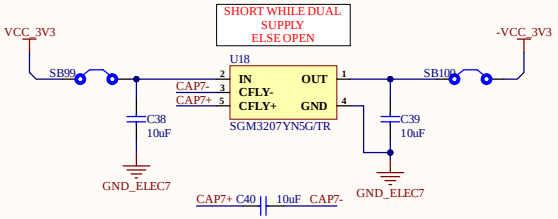


DECOUPLING CAPS

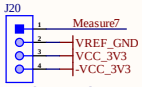


Place these near AD8221ARZ-R7

DUAL SUPPLY IC

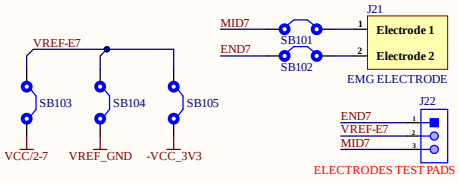


TEST PADS



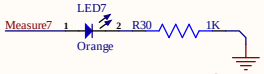
4 Pin SMD Wire Connector

sEMG SENSOR

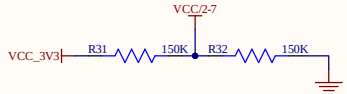


For Dual-supply, short VREF_GND SB

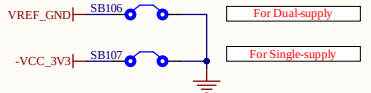
LED

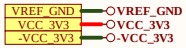


Voltage Divider



Solder Bridges



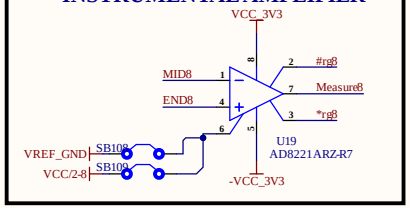




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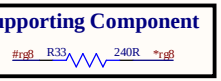
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Date: 08/02/2025	Page	Num: *
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INSTRUMENTAL AMPLIFIER

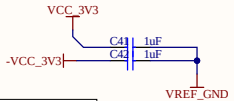


VREF input requires almost 50uA current. This would make the total current of 8 opamps equal to 400uA.

Supporting Component

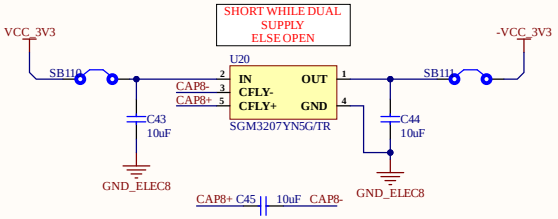


DECOUPLING CAPS

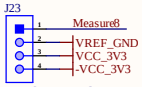


Place these near AD8221ARZ-R7

DUAL SUPPLY IC

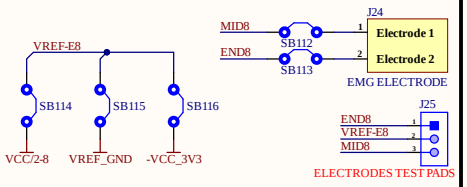


TEST PADS



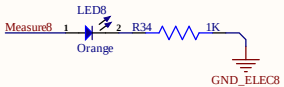
4 Pin SMD Wire Connector

sEMG SENSOR

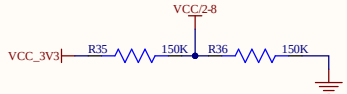


For Dual-supply, short VREF_GND SB

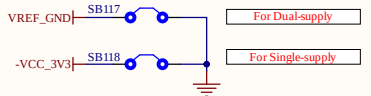
LED

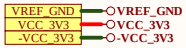


Voltage Divider



Solder Bridges



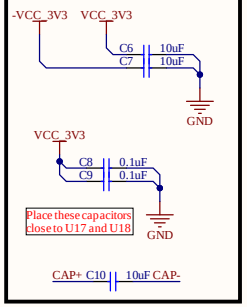




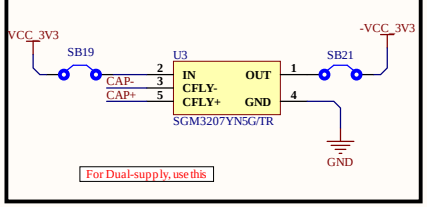
eSpark Consulting Group

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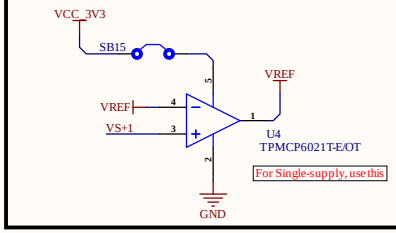
DECOUPLING CAPS



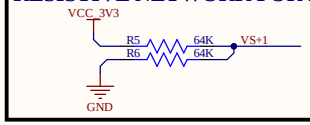
NEGATIVE RAIL



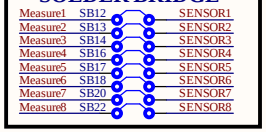
OPAMP FOR IMPEDANCE MATCHING



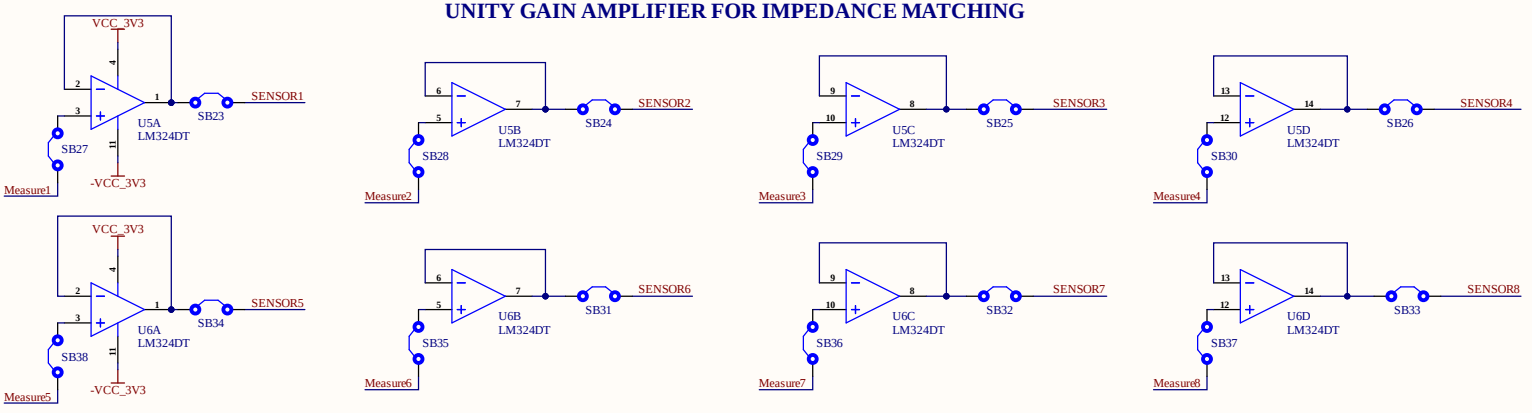
RESISTIVE NETWORK FOR 1.65V

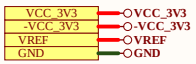


SOLDER BRIDGE

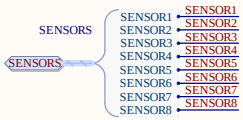


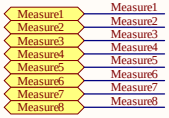
UNITY GAIN AMPLIFIER FOR IMPEDANCE MATCHING





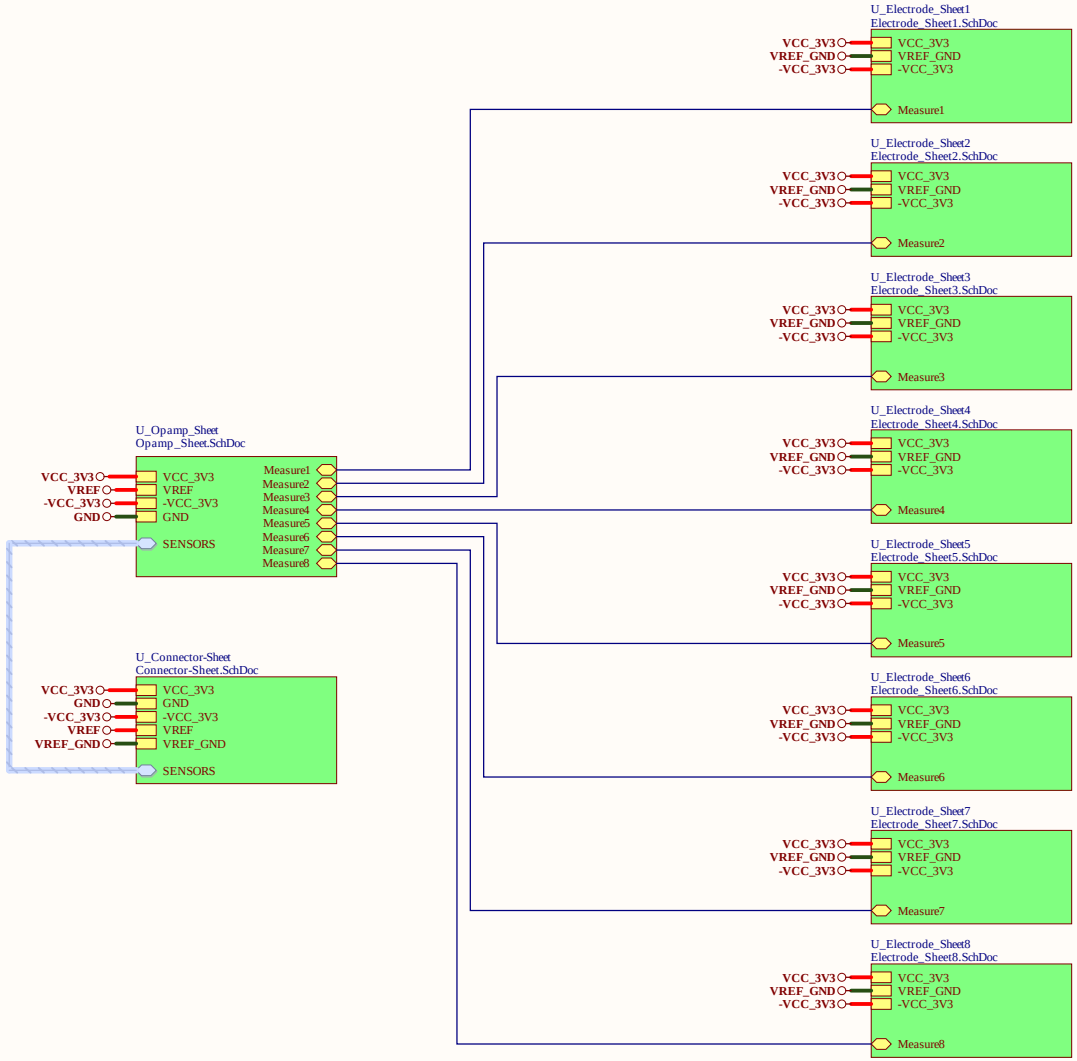
SESNOR OUT SIGNALS





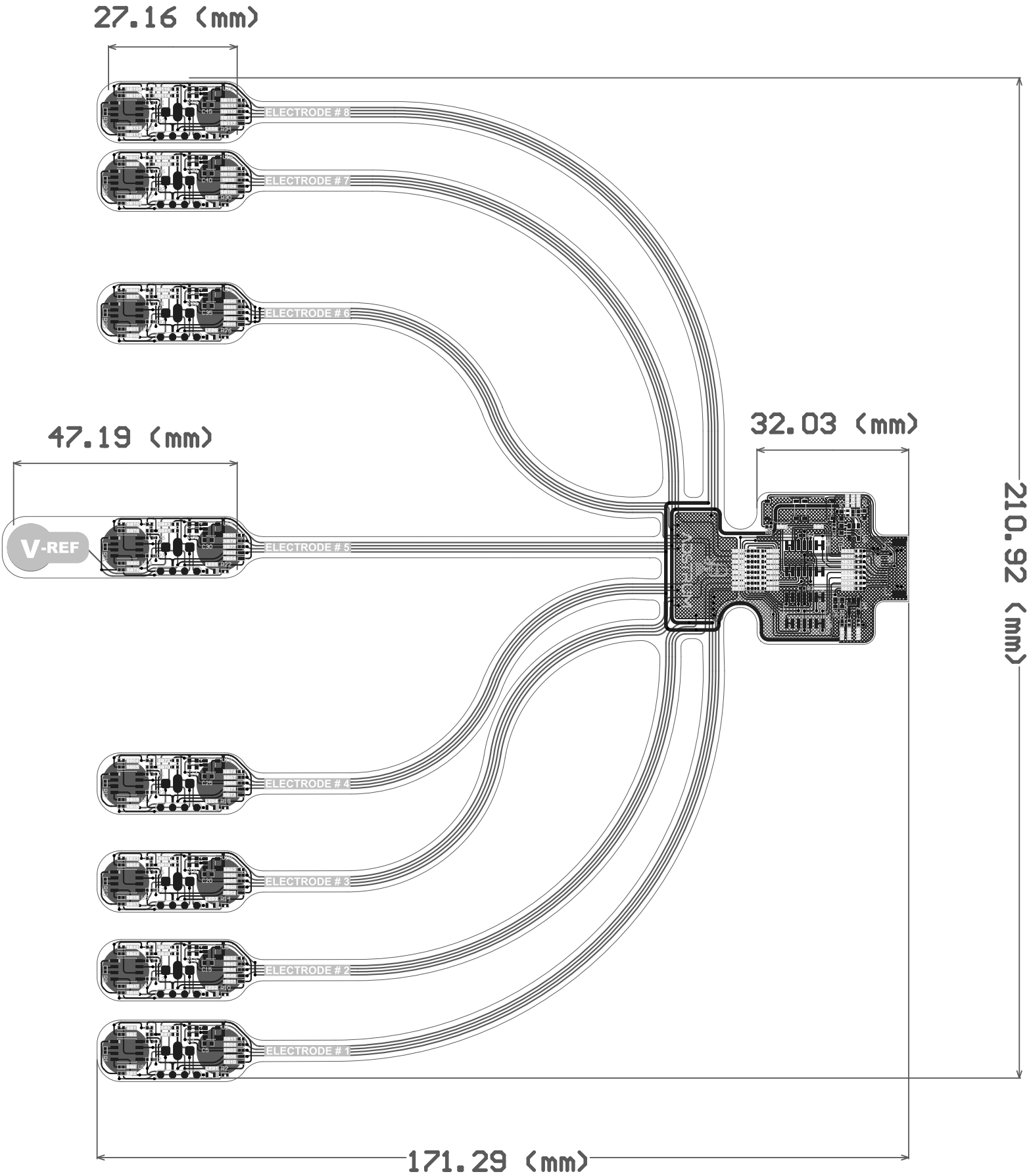
eSpark Consulting Group

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Date: 08/02/2025	Page	Num: *
Drawn By: Huzaifa Ali	Page	Total: 11


27.16 (mm)
47.19 (mm)
32.03 (mm)
210.92 (mm)
171.29 (mm)
V-REF
ELECTRODE # 8
ELECTRODE # 7
ELECTRODE # 6
ELECTRODE # 5
ELECTRODE # 4
ELECTRODE # 3
ELECTRODE # 2
ELECTRODE # 1